TrES-2b

R-1681

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_190914 · created 2026-08-02T11:39:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458738.27159 +/- 0.00068 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.04 +/- 0.043
Transit depth (Rp/Rs)^2	0.16 +/- 0.34 [%]
Orbital Inclination (inc)	85.25 +/- 2.87
Airmass coefficient 1 (a1)	0.0385 +/- 0.0027
Airmass coefficient 2 (a2)	3.0 +/- 1.65
Scatter in the residuals of the lightcurve fit is	6.73 %
Best Comparison Star	#2 - [28, 176]
Optimal Aperture	17.94
Optimal Annulus	155.48
Transit Duration (day)	0.0008 +/- 0.0016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.04 ± 0.043 vs 0.1254 (deviation 1.99σ)
Duration fit vs published	0.0008 d vs 0.0746 d (deviation 46.11σ)
Mid-time O–C	-3558.8 min
Depth SNR	0.9
Residual scatter	6.73 %

Light curve

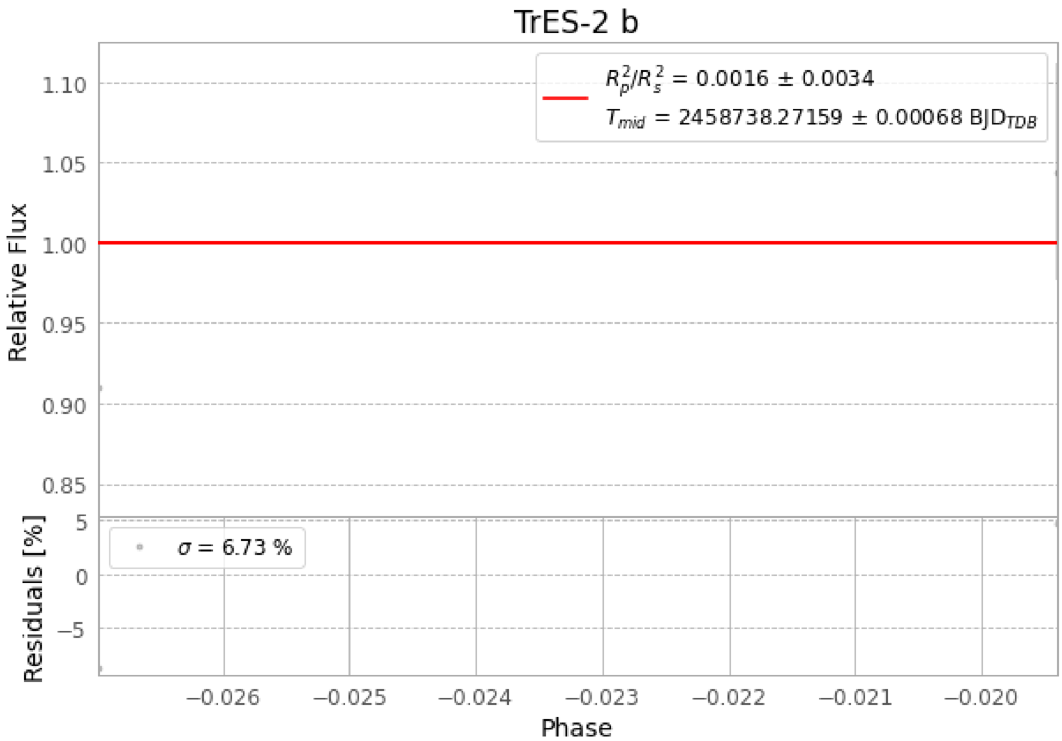

Figure 1 — FinalLightCurve_TrES-2 b_2019-09-13.png (SHA-256 80daf580c6a8c089…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [17, 214], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 f4d8120e66767039…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

70 FITS frames (46 MB), `TRES-21909140409 12.FITS` → `TRES-21909140736 12.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-190914.log (437034286efa…), FinalLightCurve_TrES-2 b_2019-09-13.png (80daf580c6a8…), FinalLightCurve_TrES-2 b_2019-09-13.csv (017a8e3483ba…)

Compute resources

Wall time 250.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)